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SUPERSEDING
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MILITARY SPECIFICATION

TRAILERS, CABLE REEL: HYDRAULICALLY DRIVEN, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers cable reel trailers that have hydraulically powered cable reel handling capabilities and that are self-propelled through a hydraulically powered landing wheel. Hydraulic power is supplied by a diesel engine driven hydraulic pump. Vehicles procured under this specification are commercial items which are required to be warranted by the manufacturer as specified in acquisition documents.

* 1.2 Classification. The trailer shall be one of the following types, as specified (see 6.2):

Single axle - 5450 kilogram (kg) (12,000 pound) payload capacity
Tandem axle - 9070 kg (20,000 pound) payload capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

*	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
	MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
	MIL-STD-1474	- Noise Limits for Army Materiel.
*	MIL-STD-1595	- Qualification of Aircraft, Missile and Aerospace Fusion Welders.
	MS 51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
	MS 51336	- Lunette-Coupler, Drawbar, Ring.
	MS 51339	- Coupler, Drawbar, Ring-Offset (Taper Shank).

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

* (Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code (BPVC)
Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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* AMERICAN WELDING SOCIETY (AWS)

AWS B2.1	- Standard for Welding Procedure and Performance Qualifications.
AWS D1.1	- Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 33125.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

J350	- Spark Arrester Test Procedure for Medium Size Engines (DOD Adopted).
* J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD Adopted).
J697	- Safety Chain of Full Trailers or Converter Dollies.
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	- Automobile, Truck, Truck Tractor, Trailer and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.13.4, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The vehicle shall be a standard commercial trailer, of a model sold to a significant number of buyers other than the Government, in the course of normal business operations.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification markings and data plates shall be in accordance with MIL-STD-1223.

* 3.1.1.2.1 Data plates. Data plates showing a schematic of all hydraulic valves, valve controls, equipment, and complete instructions for operating the hydraulic components shall be fastened on the inside of the curbside cabinet door. Data plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

* 3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor

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Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the trailer manufacturer's standard matching hardware.

* 3.1.1.5 Welders and welding. In addition to the requirements of 3.23.4, all welders employed in the fabrication of the trailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or D1.1. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

* 3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the trailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the trailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.7 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

3.1.1.8 Lunette. The coupling device specified in 3.11 shall be a lunette conforming to MS 51336. The coupling device shall also be of the offset type conforming to MS 51339 or shall provide equivalent vertical adjustment of not less than 100 mm (4 inches).

3.1.1.9 Pintle assembly. The pintle assembly on the rear tow bar, as specified in 3.12.2, shall conform to MS 51118.

* 3.1.1.10 Noise level. If the noise level at the operator's position exceeds 84 decibels (db(A)) when measured in accordance with MIL-STD-1474, a warning plate reading: "CAUTION: HEARING PROTECTION REQUIRED" shall be permanently affixed to the trailer in a conspicuous location at the operator's position. The noise level shall be determined with the vehicle stationary and all systems operating at maximum governed speed and rated loads. The plate shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.11 24-volt lighting.

* 3.1.1.11.1 Independent 24-volt dc system. When specified (see 6.2), in addition to the standard 12-volt direct current (dc) lighting system (see 3.13 through 3.13.7), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to

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cross-tav bed shall be located next to but not under the dump panel.

3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.1, except the 24-volt dc stop and turn lamps shall be combined.

* 3.1.1.11.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

* 3.1.1.11.3 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the trailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the trailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

* 3.1.1.11.4 Resistors. When an interconnected 24-volt dc system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the trailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

* 3.1.1.12 Lifting and tiedown attachments. When specified (see 6.2), the trailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the trailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments

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and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the trailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the trailer on the carrier when shipped.

* 3.1.1.13 Air transportability. When specified (see 6.2), the vehicle shall be air transportable in accordance with MIL-STD-1791 and AFSC Design Handbook DH-1-11 in types C-130, C-141 and C-5 aircraft and shall be air transportable in type C-17 aircraft. Preliminary design guidelines for the type C-17 are 3610 millimeters (mm) (142 inches) maximum height, 5180 mm (204 inches) maximum width, 20 630 mm (812 inches) maximum length, 9 degree ramp angle, and floor load capacity equal to that allowed for the type C-5 aircraft. The trailer shall be air transportable without shoring, except for contractor designed and supplied shoring for under the landing leg wheel. The trailer shall be air transportable both while coupled to a Government truck and while resting on the landing leg wheel after being successfully backed aboard the aircraft by a truck.

* 3.1.1.13.1 Removable components. Removal or relocation of mechanically attached (nonwelded, nonriveted, etc.) components with common hand tools, requiring not more than 1 manhour total to remove, relocate and tiedown; and not more than 1 manhour total to return the vehicle to its original, as opposed to reduced, configuration; shall be acceptable. The stability and mobility of the vehicle shall not be affected by reducing its configuration. Tiedowns for removed or relocated equipment shall be furnished.

* 3.1.1.13.2 Air transportability verification. Three copies of comprehensive scale drawings satisfactorily showing each stage of loading, tiedown and unloading, for each of the aircraft specified above, will be required to be furnished to the engineering offices of the procuring activity, TACOM, AMSTA-UED, for subsequent review and approval of responsible Air Force Officials at Headquarters, Aeronautical Systems Division (AFSC), Wright-Patterson Air Force Base.

* 3.1.1.13.3 Scale drawings. Scale drawings shall include plan, side and end views of the trailer and a truck and outlines of the trailer and truck negotiating required ramps and clearing required overhead door openings. As a minimum, the following information shall be shown:

MAIN FRAME
(a) Forward and rear overhang
(b) Height under front and rear overhang and at breakover angle
(c) Landing gear wheel size.

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WHEEL ASSEMBLIES
(a) Bogie articulation
(b) Number of axles
(c) Axle spacing
(d) Number of wheels per axle
(e) Tire size and inflation pressure
(f) Tire load range (ply rating)
(g) Tire footprint area.

WEIGHT/CAPACITY
(a) Weight in shipping configuration
(b) Curb weight distribution on each axle and on lunette
(c) Curb weight distribution on each axle and on landing wheel
(d) Number of tiedowns with accompanying capacity and location.

MANUFACTURER'S RATING DATA
(a) Payload rating
(b) Axle ratings
(c) Suspension ratings
(d) Landing leg rating
(e) Tire ratings.

MISCELLANEOUS
(a) Vehicle nomenclature and trailer's trade name, if any
(b) Manufacturer's model number
(c) Manufacturer's name
(d) Point of contact at the trailer manufacturer
(e) Telephone number at the trailer manufacturer.

* 3.1.1.13.4 Air transportability data plate. An air transportability data plate conforming to MIL-P-514, composition A (class 1 or 2) or composition C, shall be provided. The plate shall contain at least the following information:

(a) Side and rear silhouette views of the trailer
(b) Horizontal and vertical locations of the center of gravity of the trailer in transport condition
(c) Shipping weight
(d) Overall length, width and height
(e) Tiedown instructions for air transport in types C-130, C-141, C-5 and C-17 aircraft

* 3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of trailer furnished and in effect on the date of manufacture.

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* 3.2.2 OSHA and FMCSR regulations. The vehicle, with all furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of trailer for the same general use. See 6.1.

* 3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity.

* 3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

* 3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.7 Lubricating system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

* 3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

* 3.2.9 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.3 Performance. The trailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

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(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

* 3.3.1 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping. The trailer shall be capable of moving through an angle of not less than 60 degrees in relation to a 2590 mm (102-inch) wide towing vehicle without interference between the two vehicles. The angle shall be measured between the longitudinal axis of the trailer and the longitudinal axis of the towing vehicle at the lunette. The ability of the trailer to meet this requirement shall be demonstrated actually or graphically with a 2590 mm (102 inch) wide towing vehicle having its pintle at the extreme rear of the towing vehicle. Interference will be considered to exist if, at the 60 degree angle, any portion of the trailer 1300 mm (51 inches) or less from the lunette is forward of the pintle.

* 3.3.2 Tracking ability. The trailer shall follow the towing vehicle without exceeding the tracking deviations specified in Federal Motor Carrier Safety Regulation 393.70(a).

* 3.3.3 Service brakes. The service brakes of the trailer in conjunction with the towing vehicle (truck-trailer combination), under all conditions of loading, shall control, decelerate and stop the vehicle combination in accordance with Federal Motor Carrier Safety Regulation 393.52.

3.3.4 Automatic actuation. The trailer shall be equipped with an automatically actuated device to apply the trailer brakes upon breakaway from the towing vehicle. The breakaway system shall conform to Federal Motor Carrier Safety Regulation 393.43.

3.3.5 Lateral stability. The trailer shall maintain lateral stability during static and dynamic loadings with no evidence of any deformation which interferes with the smooth operation of the hydraulic system.

* 3.3.6 Lifting system. When the specified payload is located over the cable reel spindle, the hydraulic elevating mechanism shall hold the loaded cable reel spindle in the vertical carrying position without evidence of leakage or part failure. Further, the mechanism shall hold the loaded spindle in position even in the event of hydraulic system failure. The hydraulic system shall be capable of holding the specified payload located on the cable reel spindle for 8 hours and evidence of settling shall not exceed 25 millimeters per hour (mm/h) (1 inch per hour) vertical settling. Settling shall be measured from the top of spindle to the chassis main frame members. Hydraulic fluid shall be stabilized at ambient air temperature before testing.

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3.3.7 Line pull capacity. The trailer reel driving mechanism shall have a single line pull capacity of not less than 9070 kg (20,000 pounds).

* 3.4 Payload and payload distribution. The rated payload capacity shall be not less than 5450 kg (12,000 pounds) for single axle trailers and 9070 kg (20,000 pounds) for tandem axle trailers. The imposed loading on the axle(s), springs and tires, measured at the ground, shall be determined with the rated payload located on the cable reel spindle in travel position.

3.5 Dimensions, clearance and reel sizes.

* 3.5.1 Height. The overall height of the trailer measured from ground level when loaded with any reel within the size range specified in 3.5.4, shall be not more than the greater of:

(a) The reel diameter plus not more than 510 mm (20 inches)
(b) 1830 mm (72 inches).

* 3.5.2 Width. The overall width of the trailer, exclusive of tires, shall be not more than 2440 mm (96 inches). The width over the tires shall not exceed 2540 mm (100 inches).

* 3.5.3 Ground clearance. The ground clearance of the trailer, with a mounted 2740 mm (108 inch) diameter cable reel loaded as specified in 3.4, shall be not less than 360 mm (14 inches).

* 3.5.4 Reel diameter. The trailer shall be capable of handling, driving and transporting cable reels from 1270 mm (50 inches) to 2740 mm (108 inches) in diameter or when specified (see 6.2), to 4270 mm (168 inches) in diameter.

3.5.5 Reel width. The trailer shall be capable of handling, driving and transporting cable reels from 610 mm (24 inches) to 1420 mm (56 inches) in width.

* 3.6 Trailer configuration. The trailer shall be provided with a hydraulically operated lifting mechanism to enable reels of different sizes to be mounted without tilting the trailer. The trailer shall have a power-operated mechanism to rotate the cable reel when the reel is in travel position. The trailer shall have a driving and steerable landing wheel to enable one man to maneuver the trailer into position after detaching the trailer from the towing vehicle. The landing wheel shall be hydraulically powered and tiller steered.

3.6.1 Frame. The trailer frame members shall be adequately braced with a sufficient number of crossmembers to ensure stability. When the trailer is loaded as specified in 3.4, the maximum fiber stress in any frame member shall be not greater than 50 percent of the yield strength of the materials used. All parts of the frame shall be completely electric-arc welded. Where

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frames are spliced, joints shall be reinforced to fulfill the dynamic and static loading requirements specified herein. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

* 3.6.2 Stress analysis. A stress analysis for the trailer, loaded as specified in 3.4, shall be performed. The stress analysis shall include shear and moment diagrams and deflection calculations. Stress calculations shall include complete analysis of the crossmembers and longitudinal frame members.

3.7 Axles. Axle(s) shall be of the stub type. Axle(s) shall have a rated capacity at least equal to the load imposed, measured at the ground, with the trailer loaded as specified in 3.4.

* 3.8 Suspension system. If a suspension system is available as the manufacturer's standard or optional item, it shall be furnished. The suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, with the trailer loaded as specified in 3.4.

* 3.9 Wheels, rims and tires. Single axle trailers shall be equipped with two single wheels, tandem axle trailers with four. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size tires furnished. Rim sizes shall be the same for all wheels on the vehicle. Multi-piece rims shall not be furnished.

* 3.9.1 Tires. Steel belted radial, tubeless tires with highway tread shall be furnished. Tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the trailer loaded as specified in 3.4. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.9.2 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. The spare wheel or rim shall be the same size as those mounted on the trailer. The spare wheel or rim shall be stowed on the vehicle for shipment.

* 3.9.3 Spare tire assembly. When specified (see 6.2), an inflated spare tire mounted on a spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread design and ply rating as the tires furnished on the trailer. The spare tire assembly shall be stowed on the vehicle for shipment.

* 3.9.4 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with,

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corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.9.5. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3.9.5 Tools. When specified (see 6.2), the trailer shall be furnished with tools required for exchanging a mounted tire assembly, and shall include at least a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with a tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel assembly.

3.10 Brakes.

* 3.10.1 Service brakes, single axle trailers. Single axle trailers shall be equipped with a 12-volt electric brake system wired in accordance with 3.13.4 through 3.13.7. Brake linings shall be asbestos free. The system shall include a manual controller kit for mounting on the dash of the towing vehicle and a controller kit for synchronizing the trailer electric brakes with the brakes of the towing vehicle. Unless otherwise specified, the synchronizing controller shall be for use with hydraulic brake equipped towing vehicles. When specified (see 6.2), the synchronizing controller shall be for use with air brake equipped towing vehicles. The kits shall be complete with all necessary parts and installation instructions. The kits shall be boxed, marked as to contents and securely attached to the trailer for shipment.

* 3.10.2 Service brakes, tandem axle trailers. Service brakes on tandem axle trailers shall be of the full air type. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.50 and 393.52. Brake linings shall be of nonasbestos material. The brake system shall include:

(a) Piping
(b) Hose connections
(c) Gladhands with cover assembly
(d) Air reservoirs
(e) Automatic slack adjusters
(f) Relay emergency valve and
(g) All other associated parts and connections required for a complete air brake system.

Location of air fittings shall be in accordance with SAE J702. Air fittings shall be mounted in a recessed location. Front air hose connections shall be provided with gladhands, spring loaded dust covers, or dummy gladhands

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equipped with security chains or cables. Gladhands shall conform to SAE J318. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims.

* 3.10.3 Parking brakes, tandem axle trailers. Tandem axle trailers shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the trailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. With no additional application, the parking brakes shall hold the trailer, with rated payload, on an 10 percent incline with the trailer headed first up and then down the incline. The parking brakes shall be automatically applied upon disconnection of the supply (emergency) air line and under emergency braking conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.10.4 Chock blocks and cradles. Two pairs of wood, plastic, or steel chock blocks with corrosion-resistant captive chains or cables and two cradles shall be installed on the trailer. The hook of each captive chain or cable shall be secured to the chock block cradle. The captive chains or cables shall be of a length sufficient to be wrapped around the cradle to hold the chock blocks in place during travel and to allow the wheels to be chocked in the front and the rear. The chock blocks shall be of sufficient size to prevent trailer movement during power reeling operations, while parked headed up or down a 20 percent slope with full rated payload.

* 3.11 Drawbar. The drawbar shall be of all steel construction. The drawbar coupling device shall conform to 3.1.1.8. The drawbar shall be in accordance with Federal Motor Carrier Safety Regulation 393.70(c).

* 3.11.1 Safety chains. Safety chains shall be furnished. Chains shall be in accordance with Federal Motor Carrier Safety Regulation 393.70(d) and shall conform to SAE J697. A grab hook shall be provided on the free end of both chains.

* 3.12 Landing leg with powered wheel. A retractable landing leg shall be furnished. The landing leg shall be provided with a swivel mounted wheel driven by a hydraulic motor. The wheel shall have a pneumatic tire. The landing leg shall be provided with a tiller-type steering lever control to permit the operator to swivel the wheel and tire and thus position the trailer. The powered wheel shall be capable of moving the trailer at infinitely variable speeds up to not less than 2.4 km/h (1.5 miles per hour). The landing leg wheel controls shall be located adjacent to the tiller-type steering lever. All components of the hydraulic system shall be protected or guarded and accessible for servicing. The landing leg assembly shall accept the loads imposed, measured at the ground, with the trailer loaded as

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specified in 3.4, without evidence of binding, damage, or deformation. Vertical retraction shall be my means of a hand crank. The landing leg swivel and retracting mechanisms shall be protected from exposure to foreign matter which would impair functional efficiency.

3.12.1 Support jacks. Support jacks shall be furnished at the rear of the trailer. The jacks shall be mechanically operated and provided with not less than 125 mm (5 inch) diameter swivel type pads. The jacks shall retract or fold so that the bottom of the jacks is at or above the bottom edge of the wheelhousings. The jacks shall be corrosion-resistant and shall be capable of withstanding the imposed loads, measured at the ground, when loading the cable reel onto the mandrel, and when doing overhead or underground cable work.

* 3.12.2 Rear tow bar. When specified (see 6.2), a detachable rear tow bar shall be furnished. The tow bar design shall enable personnel to assemble and detach the towbar to and from the trailer without the use of tools. Retaining chains or cables shall be furnished to secure the tow bar to the trailer. A pintle shall be furnished on the tow bar and shall conform to 3.1.1.9. Safety chain attachment devices shall be provided on the tow bar in accordance with Federal Motor Carrier Safety Regulation 393.70(d). When assembled on the trailer, the tow bar shall be structurally adequate and properly installed to enable towing a 1360 kg (3,000-pound) gross weight, two-wheel trailer.

* 3.13 Electrical system. The electrical lighting system shall be of 12-volt direct current (dc) potential. All wiring shall be in accordance with SAE J1292. All wiring exposed to possible physical damage shall be protected by the use of heavy wall thermoplastic insulated cable or conduit. Electrical wiring passing through structural members or floor plates shall be provided with grommets and shall be supported to prevent chafing and vibration.

* 3.13.1 Lighting. The lighting system shall conform to Federal Motor Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25, 393.26 (a, b, c, and d) and 393.27 through 393.33. Wiring pattern, wire color code and the coupling connector receptacle shall conform to SAE J560. All lights and reflectors shall be mounted in recessed settings or in guarded locations to provide protection from operational hazards. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or on vehicle bumpers.

3.13.2 Lighting receptacle. The vehicle shall be equipped with lighting receptacle(s) mounted in readily accessible locations. One receptacle shall be located at the front and, when a rear tow bar is required, one at the rear. The receptacles shall conform to SAE J560. All conductors shall be connected.

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* 3.13.3 Electrical cable. An electrical cable conforming to SAE J1067, with a cable plug at each end conforming to SAE J560, shall be provided. Both plugs shall be equipped with a grip for withdrawing from connector sockets. Stowage provisions for hanging the cable with the plugs pointing down shall be provided. The cable shall be not less than 910 mm (3 feet) longer than the distance from the front trailer electrical receptacle to the lunette.

* 3.13.4 Brake wiring. Electric brakes shall be wired to the auxiliary circuit of the SAE J560 seven-conductor electrical receptacle. All wiring shall be properly mounted and secured with clamps. Brake wiring shall be heavy wall thermoplastic insulated cable and shall be enclosed in conduit.

* 3.13.5 Breakaway battery. When electric brakes are furnished, a 12-volt battery shall be installed to operate the electric brakes in the event of a breakaway from the towing vehicle. The battery shall be maintenance-free type having the maintenance-free characteristics listed in W-B-131. The reserve capacity rating measured in accordance with SAE J537 shall be not less than 35 minutes. The cold cranking rating at minus 18 degrees Celsius (^{o}C) (0 degrees Fahrenheit (^{o}F)) measured in accordance with SAE J537 shall be not less than 160 amperes. A weatherproof battery box shall be furnished.

* 3.13.6 Battery recharging circuit. A battery recharging circuit shall be wired through the first pin of the four-conductor electrical receptacle specified in 3.13.7. The second pin shall be return ground to the towing vehicle. A diode shall be installed in the system so that the electric current will not flow from the trailer battery to the towing vehicle system.

* 3.13.7 Breakaway battery wiring. The breakaway battery shall be wired to two conductors of a four-conductor electrical receptacle, Midland Model 23400 or equal. A second identical receptacle for mounting on a towing vehicle shall be furnished. The receptacles shall be polarized and weather-proof. A connecting cable, each end equipped with a mating plug, Midland Model 23403 or equal, shall be furnished. The cable shall be not less than 910 mm (3 feet) longer than the distance from the four-conductor receptacle to the lunette.

* 3.14 Rear end protection. Rear end protection in accordance with Federal Motor Carrier Safety Regulation 393.86 shall be furnished. When rear end protection interferes with loading or unloading of cable reels or pay out or recovery of a cable during overhead and underground work, bumpers at the rear of the cabinet area only may be substituted.

* 3.15 Mud flaps. Rubber mud flaps shall be installed to the rear of the wheels and shall conform to SAE J682. A metal strip not less than 3.2 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or

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bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick, shall be furnished at each bolt.

* 3.16 Wheelhousings. Wheelhousings constructed of sheet metal shall be provided. The wheelhousings shall include fenders, inside backing plates, steps and rear storage cabinets. The top surfaces shall readily support a 115 kg (250-pound) person and shall be covered with safety steel tread plate or coating of American abrasive Metals "Epoxo" nonslip adhesive coating or equal. The wheelhousings shall prevent road splash from entering onto the operating components and on to a mounted cable reel. A curbside and streetside step shall be provided. Steps shall be of subway grating and shall be located at the front end of the wheelhousings.

* 3.16.1 Storage cabinets. Rear storage cabinets, integral with and the full height of the fenders, shall be provided. Cabinet doors shall be weatherproof and constructed to maintain fit and alinement in service. Doors shall have continuous rod- or hem-type, heavy duty hinges, with stainless steel hinge pins and noncorrosive sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts shall be constructed of, or plated with, corrosion-resistant material. Doors shall be side hung. Cabinets shall be on both the road and the curbside of the trailer. Means shall be provided to flush out and drain the cabinets and to prevent road splash from entering the cabinets.

* 3.17 Reel spindle and rotating unit. The reel spindle shall be not less than 64 mm (2.50 inches) in diameter and shall be provided with adjustable cone shaped collars or guides to accommodate cable reel widths of 610 mm to 1420 mm (24 to 56 inches). To enable leveling of the spindle when the trailer is on uneven ground, both ends of the spindle shall be capable of being hydraulically raised or lowered independently. A power winding unit shall be mounted forward of the cable reel and shall rotate the cable reel forward and reverse. A brake shall be provided on the power winding unit to prevent backlash and shall provide for free spooling of the reel. The brake shall be adjustable to control braking of the reel and shall be mechanically or electrically actuated.

3.17.1 Reel holders. The reel holders shall be enclosed in a hydraulically lifted housing and shall be designed to enable the cable reel to be loaded on the vehicle by one man without the assistance of any other mechanical device. Reel lifting housings shall accommodate at least 6 mandrel positions. The lifting housing shall be capable of being mechanically locked in travel position to prevent loads on the hydraulic system and to permit the operator to recover overhead and underground cable on a reel ready for storage. The lifting housing shall have vertical travel of not less than 360 mm (14 inches).

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* 3.18 Auxiliary power shaft. An auxiliary power shaft shall be furnished on the power winding unit for accommodating a 180 mm (7 inch) pin-on detachable capstan and a wire take-up reel. The power shaft shall be on the curb side of the trailer. Means shall be provided to engage and disengage the power shaft to and from any accessories. Provisions shall be made for stowage of a capstan and a reel in the storage cabinets.

* 3.19 Controls. Controls shall be centrally located for one-man operation. Each control shall be marked with a corrosion-resistant label or a corrosion-resistant data plate depicting the function of the control. Individual markings shall be adjacent to each control.

* 3.20 Engine. A diesel engine shall be provided. The engine shall be air-cooled, mounted forward of the cable reel, and capable of driving the hydraulically operated cable reel assembly and all hydraulic components. The engine shall be covered to prevent exposure to the elements. Engine mounting shall minimize transfer of vibration. The engine shall have sufficient output power to drive all operating components. During normal operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein. In addition to operation on conventional diesel fuel, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.20.1 Engine components. The engine shall be equipped with at least the following components:

(a) Starter, 12-volt
* (b) Air cleaner
(c) Alternator, 12-volt
(d) Battery, 12-volt, with battery box and cover. The battery shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131
* (d) Hourmeter.

3.20.2 Fuel pump. A fuel pump shall be furnished on the engine. When lifting fuel against a head of 1220 mm (4 feet), the fuel pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided.

3.20.3 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

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3.20.4 Exhaust system. A spark arrester shall be furnished. The spark arrester shall have not less than an 80 percent arresting efficiency when rated in accordance with SAE J350. Means shall be provided to vent exhaust fumes away from the location of operating personnel.

3.20.5 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

* 3.20.6 Engine fuel tank. An engine fuel tank shall be provided. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a security chain or cable. Fuel tank drain(s), located to prevent spillage of drained fuel on the trailer, shall be provided. The tank filler openings shall be provided with a protective screen and a vent. A shut-off valve shall be provided in the fuel line adjacent to the fuel tank. Means shall be provided to prevent fuel line fatigue. The fuel tank shall be marked denoting the type of fuel to be used. The fuel tank shall be provided with a means to indicate the amount of fuel in the fuel tank in one-quarter of tank capacity increments.

3.20.7 Lights. In addition to other lighting specified, lights with guards and toggle switches suitable for illuminating the hydraulic controls, reel controls, and engine control panel shall be furnished.

3.20.8 Engine control panel. The engine throttle and key operated starting switch shall be installed in a weather and waterproof cabinet on a control panel, each control identified as to its function. The cabinet shall be fitted with a side hinged cover with a spring type latch and shall provide for locking with a padlock.

3.21 Hydraulic system. The hydraulic system shall consist of the components required to perform all hydraulic operations of the trailer. Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. An automatic or manual throttle actuator shall be furnished. The throttle actuator shall increase the speed of the engine when the hydraulic pump is being loaded.

3.21.1 Hydraulic cylinders. Hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt onto the cylinders and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the hydraulically operated components shall be provided with pilot check valves.

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3.21.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal, installation, and visual inspection for failure. Hydraulic lines shall be of seamless steel hydraulic tubing or wire braid base with permanently attached fittings. Hydraulic lines shall be air and liquid tight and shall be installed to prevent vibration and chafing. Provisions shall be made for bleeding air and for draining the hydraulic system.

3.21.3 Hydraulic pump. A hydraulic pump with the necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The hydraulic pump shall be driven by the diesel engine. Chain drive shall not be used. An adjustable pressure relief valve with the range of adjustment necessary for raising the payload specified in 3.4 and for operating the hydraulic landing wheel and reel spindle mechanisms shall be provided.

* 3.21.4 Reservoir. A hydraulic fluid reservoir shall be provided. The reservoir shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 20 percent expansion space. The reservoir shall be provided with a filler opening, removable oil strainer and a vented filler cap, all located to facilitate filling the tank from a 3.78 liters (L) (1-gallon) container using a funnel. The filler cap shall have a means to prevent the fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain or cable. Provision shall be made for inspection and cleaning of the interior and for draining the reservoir. The reservoir drain shall be located so that fluid will not drain on other components of the vehicle. The reservoir shall have a data plate indicating the maximum fluid level in the reservoir with:

(a) The rated payload specified herein, and
(b) A 1420 mm (56 inch) wide, 2740 mm (108 inch) diameter cable reel in travel position with hydraulic lines and components full of fluid.

A sight level or dipstick gage shall be furnished to observe the level of fluid in the reservoir.

3.21.5 Hydraulic fluid filters. Full flow type hydraulic fluid filters shall be provided at the hydraulic pump inlet and in the return hydraulic line to the reservoir. Filters shall be accessible for periodic cleaning or replacement of the cartridge. Filters shall have sufficient capacity to allow the maximum rated pump discharge.

3.21.6 Pressure gages. A shockproof pressure gage shall be installed at the discharge port of the hydraulic pump.

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3.21.7 Safety valves. A set of safety or unloading valves shall be installed in the hydraulic system to prevent overloading and subsequent damage to the hydraulic equipment. The safety valves shall be preset so as to unload at all pressures over and above that required for the rated capacity load.

3.21.8 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system. A permanent decal or plate, located near the hydraulic system fill port, shall be furnished stating the type of hydraulic fluid to be used.

* 3.22 Servicing and adjusting. Prior to acceptance of the trailer by the Government inspector, the contractor shall service and adjust the trailer for immediate operational use. The servicing and adjustment shall include at least the following:

(a) Inflation of all tires
(b) Adjustment of brakes
(c) A check for proper functioning of all lights
* (d) A check for proper function of the engine and all hydraulic components
(e) Complete lubrication with grades of lubricants recommended for ambient air temperature at the delivery point.

* 3.23 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

* 3.23.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

* 3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not

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countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

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4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Weight and payload capacity. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the lunette and axle(s). The total imposed load distribution on the lunette and the axle(s) shall be computed using the curb weight and the specified payload on the spindle. The calculated imposed load on the rear axle(s) shall be compared to the suspension, axle and tire ratings to determine if these components are at adequate capacity to conform to contract requirements.

4.3.2 Road test. The first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table I, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table II. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

* TABLE I. Road test distance.

Number of vehicles being procured	Road test distance km (miles)
1 - 10	160 (100)
11 - 20	320 (200)
21 - 30	480 (300)
31 - 40	640 (400)
41 - 50	800 (500)
51 - 100	1600 (1000)
101 - up	8050 (5000)

* TABLE II. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

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* 4.3.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

* 4.3.2.2 Test loads. The trailer shall be tested for 90 percent of the mileage with the trailer loaded as specified in 3.4 and for 10 percent of the mileage with the trailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

* 4.3.2.3 Road test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new trailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test trailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

* 4.3.4 Brake tests. Brake tests to determine conformance to requirements shall be conducted during the road test. The brake tests shall be conducted with the fully loaded trailer coupled to a truck. The truck shall have a

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gross vehicle weight rating (GVWR) not exceeding that specified in table III. The truck shall be without excessive brake capacity. GVWR shall be the maximum rating on the truck only and not necessarily the actual maximum GVW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 12 200 mm (40 feet) from a speed of 32 km/h (20 mph) shall be cause for rejection.

* TABLE III. Maximum truck characteristics.

Type of trailer	Max. truck GVWR	
	kg	(pounds)
Single axle	7250	(16,000)
Tandem axle	10 900	(24,000)

* 4.3.5 Parking brake test. To test parking brakes, the trailer, with rated payload, shall be placed on a grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.10.3. These tests shall be conducted with the trailer disconnected from the towing vehicle and resting on the landing leg.

4.3.6 Loading. Loading of a cable reel equal in weight to the required payload capacity of the trailer shall be performed by one person.

4.3.7 Positioning of vehicle with landing wheel. Positioning of the vehicle with specified payload shall be performed by one person.

* 4.3.8 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an innerconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°C (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

* 4.3.9 Lifting and tiedown test. Lifting and tiedown provisions, markings, and the connecting structural members of the vehicle shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions and the connecting structural members shall meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

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* 4.3.10 Air transportability verification. The vehicle shall be inspected to determine that it conforms to the contractor's air transportability drawings and data as submitted to the Government for transportability approval. As a minimum, the following angles, dimensions and descriptions shall be checked against the Government approved contractor's drawings and data:

(a) Angle of approach
(b) Ramp breakover angle
(c) Angle of departure
(d) Height, longitudinal location and identification of highest component on trailer
(e) Dimensions and locations of any significant projections on trailer
(f) Distribution of curb weight on each axle and lunette
(g) Distribution of curb weight on each axle and landing leg
(h) Front overhang
(i) Rear overhang
(j) Articulation of rear suspension, unloaded, each axle (curb weight)
(k) Rear axle spacing.

* 4.4 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

* 4.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.6 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required has been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for general non-tactical use by the Government for transporting cable reels, and for overhead and underground cable operations.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
* (b) Type of cable reel trailer required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
* (d) Identification of appropriate military service for painting color (see 3.1.1.1).
* (e) Identification of appropriate military service for marking (see 3.1.1.2).
(f) Independent 24-volt dc system, if required (see 3.1.1.11.1).
* (g) Interconnected 24-volt dc system, if required (see 3.1.1.11.2).
* (h) Lifting and tiedown attachments, if required (see 3.1.1.12).
* (i) Air transportability, if required (see 3.1.1.13).
* (j) Capability of handling reels up to 4270 mm (168 inches) in diameter, if required (see 3.5.4).
* (k) Spare wheel or rim, if required (see 3.9.2).
* (l) Spare tire assembly, if required (see 3.9.3).
* (m) Spare tire carrier, if required (see 3.9.4).
(n) Tools, if required (see 3.9.5).
(o) A truck air brake synchronizing controller for the trailer electric brakes, if required, in lieu of a truck hydraulic brake synchronizing controller (see 3.10.1).
* (p) Rear towbar with pintle, if required (see 3.12.2).
* (q) Satisfactory operation on JP-8 fuel, if required (see 3.20).
(r) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.21.8).
(s) Hydraulic fluid conforming to MIL-H-46170, if required (see 3.21.8).

6.3 Subject term (key word) listing.

Auxiliary diesel engine
Cable reel
Hydraulic system
* Non-tactical vehicle (NTV)
Powered landing wheel
Road test
Welder's qualifications.

6.4 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made.

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Custodians:
Army - AT
Navy - YD
Air Force - 99

Review activity:
Air Force - 84

User activity:
Navy - MC

Preparing activity:
Army - AT

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-62072C	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
Trailer, Cable Reel: Hydraulically Driven, Modified Commercial

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code) US Army Tank Automotive Command Warren, Michigan 48397-5000 AMSTA-NED	d. TELEPHONE (Include Area Code) (1) Commercial 313-574-6407 (2) AUTOVON 786-6407 (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
c. ADDRESS (Include Zip Code)	d. IF YOU DO NOT RECEIVE A REPLY WITHIN 30 DAYS CONTACT

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